



Effect of elevated incubation temperatures on learning and brain anatomy of hatchling and juvenile lizards

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Abstract

Global warming is a major threat to reptiles because temperature strongly affects their development. High incubation temperatures reduce hatchling body size and physiological performance; however, its effects on brain development and learning abilities are less well understood. In particular, it remains unclear if the effects of elevated temperatures on learning are restricted to hatchlings or instead will persist later in life. To address this gap, we examined the effect of ‘current’ and ‘future’ (end-of-century, +4 °C) incubation temperatures on hatchling and juvenile geckos *Amalosia lesueurii*, to test: (1) if elevated temperatures affect hatchling learning ability; (2) if the effects on learning persist in juvenile lizards, and (3) if and how elevated temperatures affect hatchling and juvenile brain anatomy and neuronal count. We found that fewer future-incubated hatchlings succeeded in the learning tasks. Nonetheless, the successful ones needed fewer trials to learn compared to current-incubated hatchlings, possibly due to a higher motivation. Reduced learning ability was still observed at the juvenile stage, but it did not differ between treatments due to a reduced cognitive performance of current-incubated juveniles. Future-incubated hatchlings had a smaller telencephalon, but this pattern was not found in juveniles. Neuron number and density in hatchlings or juveniles from both treatments were not different. Our results suggest that global warming will affect hatchling survival in the wild but it remains unclear if future-incubated lizards could compensate for the harmful effects of elevated temperatures. Further testing beyond the laboratory is required to understand whether phenotypic plasticity in lizards is sufficient to track global warming.

Keywords Cognition · Behavioural flexibility · Neurons · Isotropic fractionation · Climate change

Introduction

Climate change is a major driver of global biodiversity loss (Selwood et al. 2015), causing widespread changes in phenology, new species interactions, and population declines (Hughes 2000; Sinervo et al. 2010). When faced with challenging conditions, species will most likely respond via four mechanisms: (i) shifting of distribution ranges to maintain appropriate niches; (ii) genetic adaptation to occupy novel niches; (iii) plastic physiological and/or behavioural adjustments; or (iv) extinction (Visser 2008; Gilbert and Miles 2017). Species with low dispersal ability, such as reptiles, are more likely to respond through adaptation or behavioural and physiological plasticity (MacLean and Beissinger 2017), although extinction is also observed (Thomas et al. 2004; Sinervo et al. 2010). Reptiles are particularly vulnerable to climate change because temperature directly affects their development, physiology, and behaviour (Huey 1982;

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Angilletta et al. 2002). Incubation temperature strongly influences embryonic development in chelonians (Booth 2000), crocodilians (Webb and Cooper-Preston 1989) and squamates (Deeming and Ferguson 1991), with important consequences for the fitness and survival of hatchlings (Elphick and Shine 1998; Burger 1998). Because embryos cannot modify their body temperature to cope with environmental challenges, the incubation stage is considered particularly vulnerable to environmental stress, namely climate change (Sinervo et al. 2010; Levy et al. 2015). The frequency and duration of heatwaves is predicted to increase in the coming decades (Cowan et al. 2014; Dowdy et al. 2015), and consequently some reptile species may experience higher temperatures in their nests (Telemeco et al. 2009; Levy et al. 2015). To assess if such higher temperatures at the vulnerable incubation stage pose a threat to lizard populations, it is crucial to fully understand the thermal sensitivity and phenotypic responses of reptiles to higher incubation temperatures (Levy et al. 2015).

The effects of elevated incubation temperatures on hatchling phenotype can be species-specific, yet there is a general consensus that elevated temperatures have a negative impact on hatchlings' body size, growth, and physiological performance [reviewed by (Deeming 2004; Noble et al. 2018)]. Additionally, recent studies have started to explore how elevated temperatures affect behavioural and brain anatomical traits that presumably determine fitness, yet results so far have yielded both negative and positive effects. For instance, learning ability is negatively affected by high incubation temperatures in the velvet gecko, *Amalosia lesueurii* (Dayananda and Webb 2017; Abayarathna and Webb 2020), but positively affected in three-lined skinks, *Bassiana duperreyi* (Amiel and Shine 2012; Amiel et al. 2014; Clark et al. 2014). Furthermore, incubation temperature may also influence the development of brain regions associated with learning and memory, such as the medial cortex in the forebrain (Amiel et al. 2017; Beltrán et al. 2021), which is the reptilian-analogue to the hippocampus of birds and mammals (Northcutt 1978; Laurent et al. 2016). In three-lined skinks, hatchlings reared under elevated incubation temperatures had a smaller medial cortex, but estimates from 2D histology suggest they have higher neuron densities (although this method can be unreliable for counting neurons) (Amiel et al. 2017). In several lizard species, neurons generated after hatching can migrate to the medial cortex (Lopez-Garcia et al. 1988; Font et al. 2001; McDonald and Vickaryous 2018), suggesting that the brain of hatchlings may have the potential to compensate for any negative effects of incubation temperature. While providing evidence for a role of elevated incubation temperatures in generating phenotypic changes in behaviour and brain anatomy, the results of these studies do not yet offer a clear and unambiguous picture on how reptiles will respond to elevated temperatures (Beltrán

et al. 2021). Moreover, studies thus far have focussed on short-term effects of elevated temperature on hatchlings. Therefore, it remains largely understudied if cognitive and/or brain plasticity induced by elevated temperature during early ontogeny persists in juvenile phenotypes, or if effects are transient and reversible (but see Siviter et al. 2017 on social cognition of adult bearded dragons).

To address this empirical gap, we examined the effect of elevated incubation temperatures on the learning ability and brain anatomy of the velvet gecko, *Amalosia lesueurii* (Duméril and Bibron 1836). Our aims were to test (1) whether high incubation temperatures influenced associative cognitive abilities of hatchling lizards; (2) if any observed effects persisted in juvenile lizards; and (3) if elevated temperatures affected hatchling and juvenile brain anatomy and neuronal count, particularly in the telencephalon which is linked to cognitive functions. Cognitive abilities were tested with two assays—associative learning and reversal learning. Associative learning tasks are used to test whether individuals learn a simple cue-response association that under natural conditions could benefit them in a wide range of contexts, from finding food, suitable mates, or avoiding predators (Morand-Ferron 2017). Reversal learning tasks, commonly used to test for behavioural flexibility (Leal and Powell 2012; Szabo et al. 2018), are considered more cognitively demanding because animals need to inhibit their previously learnt response, and form a new association with the previously unrewarded stimulus (Sutherland and Mackintosh 2016).

The ability to learn a reversal is thus an indicator of an animal's ability to adjust to changing environmental conditions. Furthermore, rapid and flexible learning might improve competitive abilities and survival of animals in the wild (Sol et al. 2008; Morand-Ferron et al. 2016; Henke-von der Malsburg et al. 2020). For instance, individuals with greater learning abilities and behavioural flexibility cope better with novel challenges (Dukas 1998) and often respond more successfully to environmental stochasticity (Day et al. 1999; Damas-Moreira et al. 2018). For brain anatomy comparisons, we quantified the number of neurons in the forebrain and midbrain regions which are important neural correlates of cognitive abilities (Northcutt 1978; Rodríguez et al. 2002) using the isotropic fractionator, a method to quantify neuron numbers quickly and accurately (Herculano-Houzel and Lent 2005).

We predicted that elevated temperatures would produce gecko hatchlings with reduced learning ability, behavioural flexibility, and lower neuron density, particularly in the telencephalon (Northcutt 1978; Amiel et al. 2017). Additionally, we predicted that elevated temperatures would generate changes that persisted in juvenile lizards since early life conditions are often critical and can generate persistent phenotypic changes (Elphick and Shine 1998; Noble et al. 2018).

On the other hand, given that neurogenesis has been reported in several brain regions of post-hatching lizards (Font et al. 2001; Kverková et al. 2020), it is also possible that effects on brain and/or behaviour in juvenile lizards are reversible, and therefore transient.

Materials and methods

Collection of females and egg incubation treatments

During the spring (October–November) of 2017 and 2018, we collected 46 gravid female geckos (*Amalosia lesueurii*) in Mellong (33°06' S, 150°41' E), Nowra (34°56' S, 150°22' E), and Dharawal (34°12' S, 150°52' E) in the state of New South Wales, Australia. Animals were captured by hand and transported to the laboratory in less than 48 h. Each female was housed in a ventilated plastic enclosure (215 L × 152 W × 95 H mm) with moist vermiculite (water potential of –200 kPa) to a depth of 3 cm and a heating cable to heat one side of each enclosure up to approximately 32 °C. Each cage contained a circular black plastic shelter (100 D × 50 mm H) and a plastic water dish (62 D × 14 H mm). Females were each fed six crickets (~5 mm length) three times a week. Crickets were dusted with vitamins (aristopet Repti-vite®) and calcium (URS Ultimate Calcium®) once per week. Females were kept in these conditions until oviposition and then released at their site of capture.

Each egg was weighed (± 0.01 g), placed in moist vermiculite in a 100 mL glass jar, and sealed with cling wrap to prevent water loss. Glass jars and vermiculite were autoclaved before use to reduce the risk of microbial growth. Velvet geckos lay two eggs per clutch, so we randomly allocated one egg from each clutch to each incubation treatment. We used programmable cycling incubators (Labwit ZXSR-1160 BOD; Australia) to produce two experimental incubation treatments. The first treatment (hereafter referred to as 'current') mimicked the mean temperature and thermal range of communal nests under current conditions (mean = 24.5 °C, range 10–33 °C, Supplementary Information 1.1; (Dayananda et al. 2016). The second treatment (hereafter referred to as 'future') was set to have a mean increase in temperature of ~4 °C (mean = 28.5 °C; range = 14–37 °C, Supplementary Information 1.1; (Dayananda et al. 2016), which corresponds to the predicted increase in spring and summer temperatures by 2100 in south-eastern NSW (Dowdy et al. 2015). Predicted air temperature scenarios are ecologically relevant since daily temperatures recorded inside reptile nests, including our gecko species, show a strong positive correlation with daily air temperatures (Cuartas-Villa and Webb 2021). Temperature inside the incubators was

recorded every 30 min using Thermocron iButton® loggers (DSG1921G, ± 0.5 °C, Maxim Integrated Products; USA).

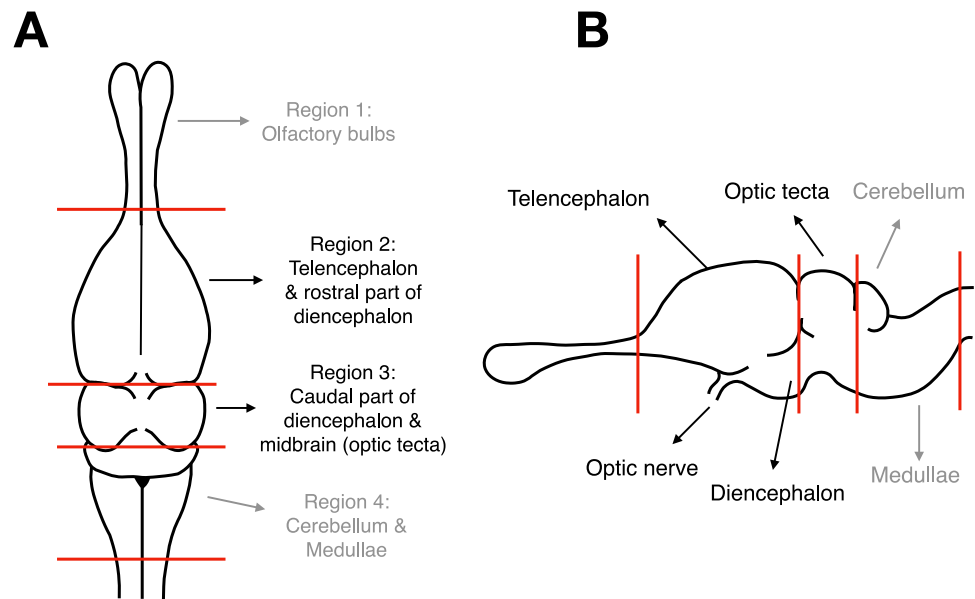
Upon hatching, we measured the lizard's weight (± 0.01 g) using a digital scale and snout-vent length (SVL ± 0.5 mm) using digital callipers. Then, each hatchling was transferred to a ventilated plastic enclosure (320 × 220 × 140 mm) with white paper as substrate, a water dish (62 D × 14 mm H), and a PVC tube (100 L × 40 W mm) cut in half-lengthwise as a refuge. Animals were housed in a room with a thermal gradient of 21–34 °C and a 12L:12D light cycle (lights on 0630–1830 h). We fed hatchlings five pin-head crickets (~3 mm length) each, three times per week. Crickets were dusted with vitamins and calcium twice per week.

Hatchlings—brain anatomy and learning tasks

To quantify the effect of incubation temperatures on the brain anatomy of newborn lizards, we euthanized and removed the brains of one-week old hatchlings (current, $n = 12$; future, $n = 11$). Animals were euthanized with an intra-peritoneal injection of 0.2 mL of sodium pentobarbital (Lethabarb® diluted 1:1 in ringers solution at 100 mg/kg). Brains were weighed, fixed in 4% phosphate-buffered paraformaldehyde for 20–25 days and then dissected (Fig. 1).

Brains were dissected using consistent anatomical landmarks as criteria; however, due to their small size, some functional regions could not be clearly separated. Consequently, we separated the brain into the four following regions: Region 1 included the olfactory bulbs and tract; Region 2 included the telencephalon and the more rostral part of the diencephalon; Region 3 included the most caudal part of the diencephalon (hypothalamus) and the midbrain (optic tecta); and Region 4 included the hindbrain (medullae & cerebellum regions). Because Regions 1 and 4 often did not yield enough tissue for neuron quantification, we evaluated Regions 2 and 3 only. These two regions were weighed (± 0.0001 g) and then analysed using isotropic fractionation. This method is a relatively quick and unbiased non-stereological way to count neuronal and non-neuronal cells from suspensions of free nuclei derived from brain tissue homogenates that yields results similar to those obtained with stereology (Herculano-Houzel and Lent 2005). For a detailed description of the protocol see (Herculano-Houzel and Lent 2005). Briefly, brain regions were dissociated into a suspension of nuclei using a 1 ml glass Tenbroeck tissue homogenizer. The homogenate was collected, centrifuged (6 min at 5000 rpm), and the pelleted nuclei were then suspended in PBS 0.1 M containing 1% 4',6-diamidino-2-phenylindole dihydrochloride (DAPI; Molecular Probes; Eugene, Oregon, USA). DAPI is a DNA-specific dye that highlights the nuclei under ultraviolet illumination. The DAPI-stained nuclear suspension was then quantified in a haemocytometer using 10 μ l aliquots. The suspension was considered homogeneous

Fig. 1 General anatomy of the lizard brain from the (A) dorsal and (B) lateral views, highlighting the regions analysed in this study using the isotropic fractionator: region 2 including the telencephalon and the more rostral part of the diencephalon, and Region 3 comprising the most caudal part of the diencephalon and the optic tecta



when counts varied by <15% across four aliquots, after which nuclear density was determined by averaging the counts from the four samples. Finally, the total number of cells in the original tissue was estimated by multiplying the mean nuclear density by total suspension volume. To estimate the total number of neurons, a 200 µl aliquot was removed from the nuclear suspension and immunoreacted for neuronal nuclear antigen (NeuN), which is expressed in all of the nuclei of most neuronal cell types but not in non-neuronal cells (Mullen et al., 1992). Nuclei were centrifuged (6 min at 7500 rpm), resuspended in a 0.2 M solution of boric acid pH 9.0, and heated for 1 h at 75 °C for epitope retrieval. Then, nuclei were again collected by centrifugation, washed in PBS 0.1 M, and incubated overnight at 4 °C with Cy3-conjugated polyclonal rabbit anti-NeuN IgG (1:200 in PBS 0.1 M; Millipore Corp.; Burlington, Massachusetts, USA). After being washed in PBS, nuclei were then suspended in a small volume of PBS for counting under the fluorescence microscope. To avoid counting bias, we registered the presence or absence of NeuN immunoreactivity for all of the DAPI-stained cells in each of the central fields of the hemocytometer chamber until at least 500 DAPI-stained cells had been examined. The proportion of NeuN-positive cells was determined separately for each region of interest and multiplied by the total number of cells to yield total number of neurons in each region. Finally, the total number of ‘non-neuronal cells’ was calculated by subtracting the number of NeuN-stained cells from the total number of cells.

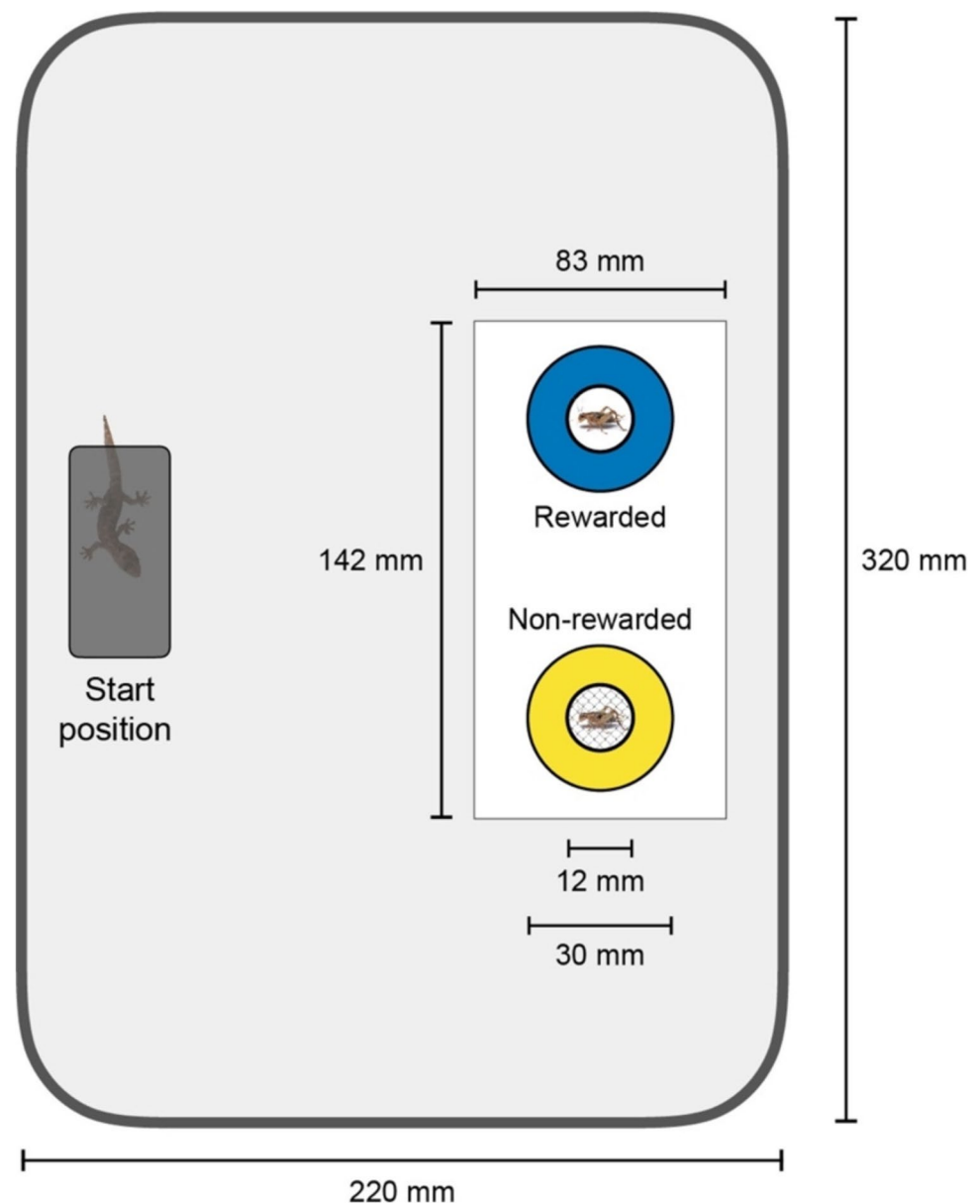
Due to technical constraints, our sample size for cell count analyses was 8 current-incubated and 11 future-incubated hatchlings.

At two weeks of age, the remaining hatchling geckos (current, $n = 16$; future, $n = 9$) were tested on a well-established

associative learning and reversal learning task (approaching a correct colour to retrieve a food item, Leal and Powell 2012; Szabo et al. 2018; Damas-Moreira et al. 2018). For both tasks, the experimental device consisted of a wooden block (142 L × 83 W × 17 mm H) with two wells (25 D × 8 mm H) carved into the top. The wells were covered by two perforated lids of different colour (yellow vs. blue) (Fig. 2). The colour of the rewarded lid was randomly determined for each individual and balanced across experimental groups. The rewarded well contained a frozen-to-death cricket accessible through the hole. The incorrect/unrewarded well also contained a dead cricket to control for chemical cues, but it was not accessible to the lizard (covered with black plastic mesh, grid size <0.5 mm). To account for potential spatial bias, the position (right or left) of the reward lid was randomised over trials (but ensuring that a given position was not repeated more than three consecutive trials). At the beginning of each trial, we ensured the animal was under the refuge opposite to the test apparatus (Fig. 2). Next, the researcher removed the water dish and the refuge before placing the apparatus inside the lizard’s home enclosure. Each trial had a maximum duration of 90 min. We considered that a lizard made a choice when its snout crossed the edge of any lid. If the animal failed to participate (i.e., no choice within 90 min and hereafter referred to as null trials), the trial was repeated. We ran three trials per day between 0900 and 1700 h with at least one hour between trials. To standardize hunger motivation, each lizard received 5 crickets daily throughout the duration of the experiments, regardless of whether they were consumed during the trials (up to 3 crickets) or provided after the trials.

Trials were video recorded; from the videos, we scored the outcome (correct or incorrect) and latency (seconds) of

Fig. 2 Schematic representation of the set-up used for the learning tasks. The experimental device consisted of a wooden block with two wells. The wells were covered by two perforated lids of different colour (yellow and blue for hatchlings; white and green for juveniles). One was randomly chosen as the ‘reward’ lid and contained a dead cricket accessible through the hole. The non-rewarded well also contained a dead cricket but had a mesh obstructing the hole



the first choice made by the hatchlings. Training for the colour association task comprised a minimum of 30 choice trials for all lizards. The learning criterion consisted of 6/6 or 8/9 correct trials (binomial test probability of success < 0.03 ; (Szabo et al. 2018). Upon completing 30 choice trials of the colour association, individuals that reached learning criterion were moved on to the reversal learning task. Individuals that did not reach learning criterion continued to a maximum of 40 choice trials and were excluded from further training if they failed to learn the task. In the reversal learning task, the procedure was the same except the reward was reversed between the coloured lids, i.e., the previously rewarded colour became unrewarded and vice-versa. Each lizard ran a minimum of 30 choice trials and continued the reversal task until it reached the learning criterion, up to a maximum of

40 trials. Since this species is nocturnal, the light cycle was reversed during the experiments (lights on 1830–0630 h) and the enclosures were illuminated with red halogen light globes (28 W). Despite being nocturnal animals, geckos use cone-based colour vision to discriminate colour under very dim light (Roth and Kelber 2004).

Juveniles—brain anatomy and learning tasks

To assess if the effects of incubation temperature on cognitive abilities were transient or persistent, we re-tested the cohort of lizards (current, $n = 15$ and future, $n = 8$; one lizard was excluded from each treatment due to lack of motivation) on the colour association and reversal learning task at 10 months of age. We kept animals from each incubation

treatment in the laboratory in common-garden conditions, in ventilated plastic enclosures with a water dish and a PVC tube cut in half-lengthwise as a refuge. The task was performed in the same way as described for hatchlings but using a different set of coloured lids (green vs. white) to avoid possible interference of memory from the previously learnt association. All juveniles ran a minimum of 30 choice trials, except for one current-incubated juvenile who stopped participating in the reversal task after 21 choice trials. Data from this individual was included and analysed. Similarly, one current-incubated juvenile completed the colour association task but failed to participate in the reversal task. At the end of the experiments, all juvenile lizards were euthanized and their brains removed for anatomy studies as described above for hatchlings.

Statistical analyses

Statistical analyses were performed in R v.4.3.2 (R Development Core Team 2018) at a significance level of $\alpha = 0.05$. Exploration of the data revealed no differences in the learning performance of hatchling or juvenile lizards between years or collection sites, thus we pooled the data from 2017 and 2018. Models were generated using ‘*lme4*’ (Bates et al. 2015). Model terms were tested for significance using the ‘*Anova*’ function in ‘*car*’ (Fox and Weisberg 2019). Comparisons with null models without interaction terms were done using the ‘*anova*’ function from ‘*stats*’ (R Core Team 2020). Post-hoc pairwise tests for multiple comparisons were run using the package ‘*emmeans*’ (Lenth 2024). Model assumptions of normality and homogeneity of variance of residuals were verified visually and using Shapiro–Wilk’s tests and Levene’s test, respectively (Zuur et al. 2010).

Our methodology included a split-clutch design to minimize potential maternal effects on our results. However, despite initially including mother ID as a random effect in our models, we did not observe any significant effects. Due to sample size limitations and the risk of overfitting, we decided to exclude this variable from further analysis.

Brain anatomy

We used Gaussian linear models to test for differences in mass, number of brain cells, number and density of neurons, and number and density of non-neuronal cells between incubation treatments and age groups. All response variables were log-transformed to fit model assumptions. When comparing total brain mass, body size was fitted as a covariate to account for allometry and focus on treatment differences in relative brain size. When analysing the mass of each of the two brain regions studied here (Region 2 and Region 3, Fig. 1), the mass of the remainder of the brain was fitted as a covariate. We used remaining brain mass (total brain

mass minus each region mass) instead of total brain mass to avoid statistical confounds resulting from the inclusion of the region being analysed in the covariate. When analysing number and density of cells, brain region mass was fitted as a covariate. For all models, the interaction term incubation treatment $\times$ age was included as a fixed effect to test if incubation effects were similar between the two age groups.

Learning experiments

To test if associative and reversal learning ability differed between incubation treatments, for each learning task we compared: (i) the number of trials needed for successful learners to reach learning criterion using a generalised linear model (Poisson distribution); (ii) the probability of learners choosing the rewarded colour per trial (correct = 1, not correct = 0) using a binomial generalized linear mixed model; and (iii) choice latency (log-transformed to fit model assumptions) of all animals to measure motivation using a Gaussian linear mixed model. For models (ii) and (iii), the interaction term incubation treatment $\times$ trial was included as the fixed effect. We also included individual level random terms for slopes (trial number) and intercepts (individual identity) to account for the possible autocorrelation between successive choices. We tested the significance of the random effects with likelihood ratio tests, by comparing models which culled the intercept or slope term from our final model.

Detailed model structure, analysis code, and model results are available as Supplementary Information 1.

Although we were interested in assessing if incubation effects on learning are transient or persistent in juveniles, we analysed learning performance of hatchlings and juveniles separately because learning success at the juvenile stage was low. We therefore have higher sample size to compare performance of hatchlings if we analyse the two age groups separately. In an exploratory analysis we included the predictor ‘trials to learning criterion as hatchling’ in our analyses of juvenile associative and reversal learning, but the low sample size resulted in model overfitting (Supplementary Information 1 (3.2)). Similarly, we chose not to include brain traits as covariates in the analyses of learning performance for two main reasons, one being the fact that hatchling brain data does not correspond to the same individuals tested in the learning tasks and the other being model overfitting from low sample size.

Results

We observed a lower, but not statistically significant, hatchling success in the future-incubated group (43.5%) compared to the current-incubated group (60%) ($P = 0.098$;

Supplementary Information 1 (1.2)). Additionally, we found that current-incubated hatchlings hatched later and were slightly heavier and larger compared to future-incubated hatchlings (all $P < 0.05$), but differences disappeared at the juvenile stage (Supplementary Information 2, Table S1).

Brain anatomy in hatchlings and juveniles

Total brain mass, after accounting for allometry with body mass, was similar between incubation treatments for both hatchling and juvenile lizards (incubation treatment: $P = 0.62$; age: $P < 0.01$; incubation treatment $\times$ age: $P = 0.35$; Supplementary Information 1 (2.2)).

A significant interaction between incubation treatment and age (Fig. 3a, b; incubation treatment: $\beta = -0.03 \pm 0.02$, $t = -1.91$, $P = 0.06$; age: $P < 0.01$; incubation treatment $\times$ age: $\beta = 0.06 \pm 0.03$, $t = 2.21$, $P = 0.03$) suggested that hatchlings incubated under future temperatures had a smaller Region 2 (telencephalon and rostral part of diencephalon) relative to total brain mass, but this effect did not persist in juveniles. Nonetheless, the effect was only marginally significant and disappeared after conducting post-hoc pairwise analyses to correct for multiple comparisons (current vs. future hatchlings: $\beta = 0.04 \pm 0.02$, $t = 1.92$, $P = 0.23$; Supplementary Information 1 (2.3)).

The mass of Region 3 did not differ between incubation treatments at any age (Fig. 3b; incubation treatment: $P = 0.24$; age: $P < 0.01$; incubation treatment $\times$ age: $P = 0.23$;

Supplementary Information 1 (2.3)). The density and total number of neuronal and non-neuronal cells in each of these brain regions did not differ between incubation treatments at any age (Fig. 3c, d; all $P > 0.1$; Supplementary Information 1 (2.4)).

Learning abilities of hatchlings

In the associative learning task, only five of nine (56%) future-incubated hatchlings reached the learning criterion, compared to all 16 (100%) current-incubated hatchlings (Fisher's Exact Test, $P = 0.009$). When comparing the subset of hatchlings that succeeded in learning the task, we found that future-incubated hatchlings needed fewer trials to learn the task (Fig. 4a; incubation treatment: $\beta = -0.39 \pm 0.14$, $Z = -2.88$, $P < 0.01$; Supplementary Information 1 (3.1.1)) and improved their probability of choosing the correct option over trials at a similar rate (incubation treatment: $P = 0.08$, trial: $P < 0.01$, incubation treatment $\times$ trial: $\beta = 0.053 \pm 0.04$, $Z = -1.51$, $P = 0.13$) compared to current-incubated hatchlings. When comparing the latency of choice among all hatchlings that participated in the task, we found that hatchlings became faster in making a choice as the experiment progressed, with similar latency rates between incubation treatments (incubation treatment: $P = 0.49$; trial: $P = 0.09$; incubation treatment $\times$ trial: $P = 0.99$; Supplementary Information 1 (3.1.1.3)).

In the reversal learning task, only one of five (20%) future-incubated hatchlings reached the learning criterion, compared to 11 of 16 (69%) current-incubated hatchlings (Fisher's Exact Test, $P = 0.11$). The performance of the single future-incubated learner fell within the range of current-incubated learners, but we cannot statistically compare the two incubation treatments (Fig. 4b; Supplementary Information 1 (3.1.2)). When comparing the latency of choice among all hatchlings that participated in the reversal task, we found that future-incubated hatchlings tended to be slower in making a choice (incubation treatment: $\beta = 0.47 \pm 0.21$, $t_{17.5} = 2.22$, $P = 0.039$; trial: $\beta = -0.01 \pm 0.006$, $t = -2.04$, $P = 0.05$; Supplementary Information 1 (3.1.2.3)).

Learning abilities of juveniles

In the associative learning task, the success rate of juvenile geckos was low in both groups; four of eight (50%) future-incubated juveniles and six of 15 (40%) current-incubated juveniles reached learning criterion (Fisher's Exact Test, $P = 0.68$). Of the subset of juveniles that succeeded in learning the task, we found that future-incubated geckos needed fewer trials to attain the learning criterion (Fig. 4a; incubation treatment: $\beta = -0.33 \pm 0.14$, $Z = -2.39$, $P = 0.017$; Supplementary Information 1 (3.2.1)), similar to results in hatchlings. Juveniles from both incubation

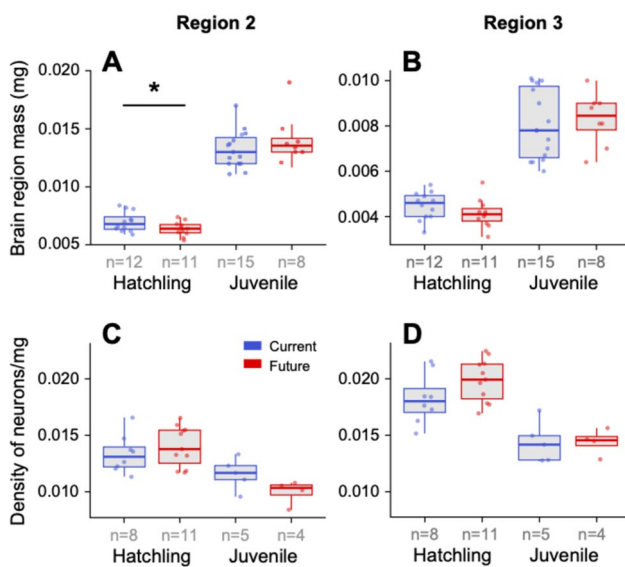
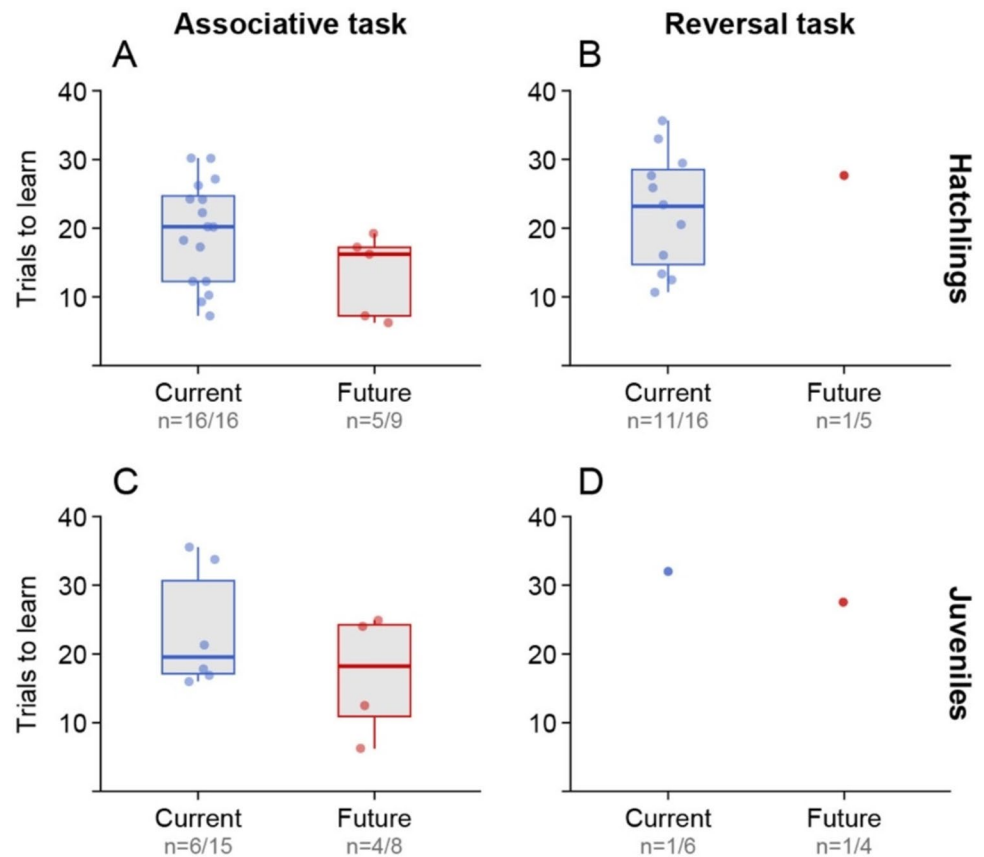


Fig. 3 Mass and neuron density of two brain regions of hatchlings and juveniles of the velvet gecko *Amalosia lesueurii*. (a–c) Region 2 comprised the telencephalon and the more rostral part of the diencephalon. (b–d) Region 3 comprised the most caudal part of the diencephalon (hypothalamus) and the midbrain (optic tecta). The number of individuals is indicated at the bottom of each panel

Fig. 4 Learning performance of the velvet gecko *Amalosia lesueurii*. Number of trials needed for hatchlings (**a–b**) and juveniles (**c–d**) to learn a colour association task (left panels) followed by a reversal learning task (right panels). The number of lizards that reached learning criterion over the total number of animals per group is given at the bottom of each panel



groups improved their probability of choosing the correct option over trials at a similar rate (incubation treatment: $P=0.21$; trial: $\beta=0.035 \pm 0.015$, $Z=2.31$, $P=0.02$; incubation treatment $\times$ trial: $P=0.25$). When comparing the latency of choice among all juveniles that participated in the task, we found that hatchlings became faster in making a choice as the experiment progressed, with similar latency rates between incubation treatments (incubation treatment: $P=0.78$; trial: $\beta=-0.01 \pm 0.008$, $t=-2.00$, $P=0.06$; incubation treatment $\times$ trial: $P=0.80$; Supplementary Information 1 (3.2.1.3)).

In the reversal learning task, only one in four (25%) future-incubated juvenile and one in five (20%) current-incubated juveniles reached learning criterion (Fisher's Exact Test, $P=0.99$). The learning speed of the two individuals was similar (31 trials future-incubated vs. 36 trials current-incubated), but we cannot statistically compare the two incubation treatments (Fig. 4d; Supplementary Information 1 (3.2.2)). When comparing the latency of choice among all juveniles that participated in the reversal task, the two groups showed similar choice latencies throughout the experiment (incubation treatment: $P=0.56$; trial: $P=0.08$; incubation treatment $\times$ trial: $P=0.29$; Supplementary Information 1 (3.2.2.3)).

Discussion

Global warming can impact the morphology and behaviour of squamate hatchlings and recent studies suggest that these effects might extend to an animal's cognitive ability. Here, we show that elevated temperatures had a negative effect on the associative learning and behavioural flexibility of hatchlings, since fewer animals succeeded in learning the two tasks compared to hatchlings from 'current' incubation temperatures. On the other hand, successful future-incubated hatchlings needed less trials to reach learning criterion. These patterns were maintained in juvenile lizards, yet current-incubated juveniles showed significantly lower learning success. Surprisingly, we found little differences in brain anatomy between incubation treatments. Future-incubated hatchlings had a smaller telencephalon and rostral part of the diencephalon, which are linked to cognitive processing, but we found no differences between juveniles. Total brain mass and density and number of neuronal cells of hatchlings or juveniles did not differ between incubation treatments.

Incubation temperature is known to influence several aspects of the morphology and behaviour of reptiles,

including body size (Elphick and Shine 1998; Booth 2000; Braña and Ji 2000), growth rate (Van Damme et al. 1992; Wapstra 2000), locomotor performance (Elphick and Shine 1998; Fisher et al. 2014; Dayananda et al. 2017), and foraging and antipredator behaviour (Burger 1991; Siviter et al. 2019). Recent studies have explored the impact of incubation temperature on lizards' learning performance, yet results so far have yielded both negative and positive effects (Amiel and Shine 2012; Clark et al. 2014; Dayananda and Webb 2017; Siviter et al. 2017; Abayarathna and Webb 2020). For instance, high incubation temperatures negatively impacted spatial learning in velvet geckos (Dayananda and Webb 2017; Abayarathna and Webb 2020) and social cognition in bearded dragons (Siviter et al. 2017), while three-lined skinks incubated at warmer temperatures performed better in spatial learning tasks (Amiel and Shine 2012; Amiel et al. 2014) compared to hatchlings incubated at lower temperatures. In the present study, we found that elevated temperatures negatively impacted the number of hatchlings of the velvet gecko *Amalosia lesueurii* reaching learning criterion in an associative learning task. However, the few animals that learnt took less trials to succeed compared to hatchlings incubated at lower temperatures. The results of our study suggest that elevated temperatures may have differential effects across individuals, potentially due to variation in other factors such as metabolism or maternal effects (Kar et al. 2022; Hoffman et al. 2023).

On one hand, we found that many individuals exposed to elevated temperatures had impaired learning, as well as reduced hatching success, and lower body weight and size. Elevated incubation temperatures are known to accelerate metabolic rates, thereby reducing the total amount of yolk material converted into tissue (osseous, muscular, and neural), ultimately hindering the embryo's development and growth (Booth 2000; Storm and Angilletta 2007). Elevated temperatures are also known to induce significant metabolic expenditure and food consumption rates in a range of ectotherms, including reptiles (Seebacher et al. 2014; Rosa et al. 2014; Kar et al. 2022). It is possible that some future-incubated lizards placed a higher value on the food reward compared to lizards from 'current' conditions, leading them to participate in the task at higher rates and, consequently, learn faster (Vila Pouca et al. 2019). In fact, the number of null trials between the future-incubated hatchlings that learnt the associative learning task (mean $\pm$ SD = 2.2 ± 4.3 trials) and those who did not (mean $\pm$ SD = 10 ± 3.8 trials) was different although not significant (Mann–Whitney–Wilcoxon Test, $W = 18$, $P = 0.06$, $N = 9$). Similarly, when comparing the latency of choice between future-incubated learners and not learners, we found that learner lizards made consistently faster choices irrespective of the trial number (learn: $P < 0.01$; trial: $P = 0.22$).

One potential reason for the contrasting positive and negative effects of incubation treatment on lizard learning performance may be that absolute temperature treatments often vary between studies, with 'hot' treatments not always representing thermal stress situations for the species; for example, the 'hot' temperature treatment used by Amiel et al. (2014) and Amiel and Shine (2012) mimicked a thermal regime experienced by natural populations at low elevation. Likewise, current studies have focused on different domains of learning (e.g. social, spatial and visual learning) complicating the unravelling of incubation temperature effects on cognition (Amiel and Shine 2012; Amiel et al. 2014; Siviter et al. 2017). For instance, high incubation temperatures had a consistent negative effect on the spatial learning of velvet geckos (Dayananda et al. 2017; Abayarathna and Webb 2020), whereas our study, testing associative learning with a food reward, found mixed results. Generally, the negative effect of elevated temperatures might have important consequences for their survival and fitness, particularly at this vulnerable life stage. In fact, hatchling velvet geckos with poorer spatial learning ability have lower chances of surviving in the wild (Dayananda and Webb 2017). However, it remains unclear whether a higher motivation to feed, and likely associated higher valuation of the food reward, by future-incubated lizards might enable individuals to compensate for the effect of climate change (Vila Pouca et al. 2019). Further work is therefore needed to disentangle effects of warm incubation treatments within the thermal range of the species versus effects of elevated temperatures associated with global warming, to fully understand how reptiles will respond to, and potentially be threatened by, future climate change.

Given the effect of incubation temperature on lizards' cognitive abilities, we expected to see an effect on brain anatomy as well. Some studies have shown that incubation temperature affects reptile brain development (Coomber et al. 1997; Crews et al. 1997; Beltrán et al. 2021), including regions associated with learning and memory (Amiel et al. 2017). Here, we also found that incubation temperature affected the brain mass of the Region 2 (telencephalon/rostral diencephalon), which includes the medial cortex—the reptilian-analogue to the hippocampus of birds and mammals (Northcutt 1978; Laurent et al. 2016), associated with learning and memory. Our results should nonetheless be carefully interpreted as the limitations encountered during dissection mean that Region 2 includes the telencephalon, parts of the optic nerve, and the more rostral portion of the diencephalon. The relationships between these brain regions and learning in reptiles are not well-documented, however they may serve different cognitive functions (Northcutt 2013; Roth et al. 2019). Our future-incubated hatchlings had a smaller telencephalon/rostral diencephalon compared to current-incubated hatchlings. This is consistent with our

findings on cognitive performance, since a large proportion of future-incubated hatchlings failed to learn both the associative learning and the reversal learning task. These results therefore further establish the telencephalon region of the reptile brain as a neural correlate of cognitive abilities (Rodríguez et al. 2002; López et al. 2003) and suggest that elevated temperatures during incubation may impact cognitive abilities, at least partially, via changes in brain region size.

In line with the findings of (Amiel et al. 2017) in three-lined skinks *Bassiana duperreyi*, we found that incubation temperature led to a decrease in the size of the telencephalon but not of the hypothalamus (a part of Region 3 in this study). This suggests a tissue-specific effect of incubation temperature on the brains of velvet geckos. However, our small sample size may have limited our ability to detect differences, as incubation temperature has been shown to induce changes in the size of the hypothalamus in lizard brains (Coomber et al. 1997; Sakata et al. 2000).

Although the correlation between brain mass, neuron number, and cognitive ability is complex, some studies highlight that examining the numbers of neurons, as the basic computational units of the brain, is essential to fully understand brain and cognitive processing (Herculano-Houzel 2017; Güntürkün et al. 2020; Kverková et al. 2022; Storks et al. 2023). In our study, we did not find an effect of incubation temperature on the number of brain cells, total number of neurons and non-neurons, nor neuron density. These results contrast with those in three-lined skinks, where high incubation temperatures led to an increase in density of neurons (Amiel et al. 2017). A lack of differences in neuron number and density in our study is particularly surprising since we did find differences in telencephalon mass between incubation treatments. Nonetheless, our sample size is small and therefore might have simply limited our ability to detect differences in cell numbers or density between treatments. The mechanisms underlying changes in the brain due to developmental temperature remains largely unexplored and could involve anatomical as well as functional changes. It is possible that hatchlings from our incubation treatments also had differences in neuron size, which has been linked to cognitive performance in birds (Freas et al. 2013). Similarly, it is possible that the effect of incubation temperature on *A. lesueurii*'s brain development occurs at a physiological level, rather than a morphological level by, for instance, increasing the metabolic capacity or the activity of dopaminergic circuits in some brain regions (Coomber et al. 1997; Dias et al. 2007). Therefore, including physiological measurements could be important in further studies.

A key question in climate change research is whether organisms will eventually overcome the impacts of the substantial changes to their environment. In animals, particularly ectotherms whose development depends on

environmental factors, it is important to know if the consequences of elevated temperatures on hatchling development will persist into adulthood and impact population viability. Our results show that, at the juvenile stage, learning performance did not differ between incubation treatments. This might initially suggest that the negative effect of temperature on hatchling cognitive abilities had disappeared. However, a closer examination reveals that the absence of differences between incubation treatments is due to a significant decline in performance of current-incubated juveniles rather than an improvement from future-incubated juveniles.

Future-incubated lizards consistently exhibited poor performance from hatchling to juveniles, suggesting they may have reached a baseline level of performance, equivalent to a random choice in learning tasks. In contrast, current-incubated lizards showed significantly poorer performance at the juvenile stage, indicating a potential detrimental effect of captivity. Existing research shows that captivity can negatively affect cognitive development in many animal groups (Crates et al. 2023), including lizards (Vardi et al. 2020); further suggesting that our results might not be representative of the situation in natural environments.

Incubation-induced phenotypic changes may play an important role in early survival. However, the relative importance of these changes in morphology and behaviour, as well as their interaction, in determining the success of hatchlings in the wild remains unclear. For instance, high incubation temperatures generate positive and negative correlations between morphology, behaviour and cognition in various lizard species (Amiel et al. 2014; Dayananda and Webb 2017; Siviter et al. 2019). It remains unclear if animals in the wild can compensate the long-lasting effects of incubation temperature on morphology and performance with improved cognitive abilities. For instance, by learning the location of safe refuges or better foraging areas to reduce their detectability to predators (Lefebvre et al. 2004; Sol et al. 2005).

In summary, our study shows that higher incubation temperatures predicted to occur in the future will negatively affect the brain mass and cognitive abilities of newborn lizards, despite few future-incubated lizards completing the tasks faster possibly due to a higher relevance of the food reward. Our findings also provide further evidence that captivity may adversely affect cognitive performance, and therefore highlights the importance of testing beyond the laboratory to understand whether phenotypic plasticity in lizards is sufficient to track global warming. The study of the effect of elevated temperatures on reptile brain anatomy and cognition offers an interesting opportunity for studies on the ecological and developmental bases of intelligence in vertebrates, but also provides valuable information on the possible effect of climate change on reptile phenotypes.

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Data availability The raw data and R script are available through the Open Science Framework at: <https://osf.io/cgq4n/>.

Declarations

Conflict of interest We declare we have no competing interest.

Ethical approval All procedures were done in accordance with the ethical standards of the Macquarie University Animal Ethics Committee, ARA No. 2017/029. Collection of animals was approved by the New South Wales National Parks and Wildlife Service, Office of Environment and Heritage (OEH; license No. SL101013 and SL101962).

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